

Project 1

Project Overview.

You will write a menu driven application that:

- a. Reads in employee and student data from a file.
- b. Stores the data in class objects and data structures.
- c. Displays the data in different ways depending on the menu choice.
- d. We will add more to our menu each week.

Project 1 Overview.

This week, we will only add Employee information from one file.

- a. Define an abstract Person class
- b. Define a concrete Employee class that inherits from Person
- c. Open a file that contains employee data
- d. Use the employee data to create a list of Employee objects
- e. Display two reports of employee data
 - a. Full employment information
 - b. Contact Information

Reports outputs are shown at the end of this document

Key data structures.

Class Person

firstName	String
lastName	String
idNumber	Int
emailAddress	String
phoneNumber	String

Class Employee inherits from Person

Class variables

roleDictionary	{001: Staff, 002: Faculty}
classificationDictionary	{001: Full-time, 002: Part-time}

Instance variables

firstName	String	//inherited from Person
lastName	String	//inherited from Person
idNumber	Int	//inherited from Person
emailAddress	String	//inherited from Person
phoneNumber	String	//inherited from Person
hireDate	date object	//See explanation below
rolePerson	string	//key from the roleDictionary - see explanation below
classificationPerson	string	//key from the classificationDictionary – see explanation below
annualSalary	Float	//Annual salary is 2 decimal places and cannot be negative

1. hireDate:

The datetime module has a date class that has year, month day

```
# import the date class from datetime
from datetime import date
```

```
#in your class, when you create a hire date
this._hireDate = date(year, month, day)
```

Note that year, month and day must be integers

See <https://www.geeksforgeeks.org/python-datetime-date-class/>

2. rolePerson

Employees will have a role at the college. For our purposes, we will only have two roles – Staff and Faculty.

Typically, a role will be represented by a code. In our case, 001 will represent Staff and 002 will represent Faculty. These relationships are kept in a dictionary that is a **class variable**

```
roleDictionary      {001: Staff, 002: Faculty}
```

If the input file show the employee is Staff, then the key value 001 should be stored in rolePerson.
If the input file show the employee is Faculty, then the key value 002 should be stored in rolePerson.
The code should allow for more roles to be added to the dictionary.

Python treats integers with leading zeros as octal, so the keys 001, and 002 should be strings.

3. classification

Employees will have a classification at the college. For our purposes, we will only have two classifications – Full-time and Part-time.

Typically, a classification will be represented by a code. In our case, 001 will represent Full-time and 002 will represent Part-time. These relationships are kept in a dictionary that is a **class variable**

```
classificationDictionary {001: Full-time, 002: Part-time}
```

If the input file show the employee is Full-time, then the key value 001 should be stored in classificationPerson.
If the input file show the employee is Part-time, then the key value 002 should be stored classificationPerson.
The code should allow for more classifications to be added to the dictionary.

Python treats integers with leading zeros as octal, so the keys 001, and 002 should be strings.

- a. Define a class Person with the following:
 - i. Inherits from Abstract Base Class ABC.
 - ii. firstName (String). May be edited.
 - iii. lastName (String). May be edited.
 - iv. idNumber (Integer - 4 digits). May **not** be edited.
 - v. emailAddress - (String). May be edited.
 - vi. phoneNumber - (String 12 characters). May be edited.

- b. Define class Employee with the following
 - i. Inherits from Person
 - ii. Has a class variable roleDictionary = {001:Staff, 002:Faculty} editable
 - iii. Has a class variable classificationDictionary = {001: Full-Time, 002:Part Time}
 - iv. Has a hireDate – May **not** be edited.
 - v. Has an instance variable rolePerson. (string), Must be a key in roleDictionary
 - vi. Has an instance variable classificationPerson (string). This should be a key in the classificationDictionary
 - vii. Has an annualSalary (Float, 2 decimal places, may **not** be negative)

Input and Processing

- a. Create a list called employeeList that will hold Employee object

- b. Write a function called getEmployees() that
 - i. Opens the employees.txt file
 - ii. Reads the file line by line
 - iii. For each line, creates an Employee object and adds it to the Employee list. Use the following link as a basic hint. <https://www.geeksforgeeks.org/how-to-create-a-list-of-object-in-python-class/>
 - iv. Notice that the text file fields for Classification and Role are values in the dictionaries, When we create our instance, for the rolePerson and classificationPerson, we should
 1. Use the respective dictionaries to look up the key for the value. See this hint: <https://www.geeksforgeeks.org/python-get-key-from-value-in-dictionary/>
 2. Store the key in the Class instance
 3. Skip adding an employee that does not have a valid role or classification
 - v. As each Employee is added, displays a message that the employee is being added

Added employee Larry Harper...
Added employee Ryan Benton...
Added employee Dave Farell...
etc...

Output

This project will be menu driven. As we add more elements to the project, we will add more elements to the menu and/or change the items in the menu. Your initial menu will look like.

Please select an option below

1. Quit
2. Display Employee Employment Information
3. Display Employee Contact Information

- a. Write a function called createMenu()
 - i. **Option 1** will have a label "Quit".
 1. When entered, it will display a goodbye message and
 2. It will exit the program.
 3. The output will look like

```
1. Quit
2. Display Employee Employment Information
3. Display Employee Contact Information
```

```
>: 1
```

Thank you for using the system.

Now exiting the program...

This should be called inside a while loop and displayed until option 1 is entered.
If an invalid option is entered, display an error message and display the menu again.

- ii. **Option 2:** will have the label "Display Employee Employment Information"
 1. When entered, will call the function displayEmployeeEmploymentInformation()
 2. The function displayEmployeeEmploymentInformation() will
 - a. Print a header line that includes
 Lastname, Firstname, ID, Email, Phone, HireDate, Classification, Role, Salary
 - b. Print the information for each employee in a nicely formatted layout. **DO NOT USE TABS.** Notice that for Classification and Role, we will have to use the keys stored in the Employee object and look up their values in the dictionary before printing them. Use this link as a guide for formatting layout. <https://www.tutorialspoint.com/how-to-align-text-strings-using-python>
 - c. The output will look like

LastName	FirstName	ID	Email	Phone	HireDate	Classification	Role	Salary
Harper	Larry	4066	Larry.Harper@email.edu	520-555-4066	5/5/2021	Full	Staff	65000.00
Benton	Ryan	4071	Ryan.Benton@email.edu	520-555-4071	10/6/2020	Full	Staff	56000.00
Farrell	Dave	4009	Dave.Farrell@email.edu	520-555-4009	10/6/2018	Part	Staff	34000.00
Stone	Matthew	4041	Matthew.Stone@email.edu	520-555-4041	8/23/2017	Full	Faculty	78000.00
Martinez	Yessenia	4004	Yessenia.Martinez@email.edu	520-555-4004	8/3/2017	Part	Staff	30000.00
Bernard	Mariah	4100	Mariah.Bernard@email.edu	520-555-4100	11/15/2016	Full	Faculty	120000.00
Day	Erin	4006	Erin.Day@email.edu	520-555-4006	10/28/2015	Full	Staff	115000.00
Oconnell	Summer	4099	Summer.Oconnell@email.edu	520-555-4099	4/12/2012	Full	Faculty	71000.00
Stevens	Regina	4079	Regina.Stevens@email.edu	520-555-4079	11/28/2011	Part	Staff	48000.00
Atkinson	Jonathon	4077	Jonathon.Atkinson@email.edu	520-555-4077	7/19/2011	Full	Faculty	48000.00
Levine	Sean	4026	Sean.Levine@email.edu	520-555-4026	10/9/2010	Part	Faculty	34000.00
Stephenson	Reginald	4083	Reginald.Stephenson@email.edu	520-555-4083	2/8/2010	Part	Staff	23000.00
Garner	Kelli	4062	Kelli.Garner@email.edu	520-555-4062	1/17/2010	Full	Faculty	58000.00
Walsh	Kirk	4042	Kirk.Walsh@email.edu	520-555-4042	4/8/2009	Full	Staff	92000.00

- iii. **Option 3** will have the label "Display Employee Contact Information"
 1. When entered, will call function displayEmployeeContactInformation()
 2. The function displayEmployeeContactInformation() will

- a. Print a header line that includes
Lastname, Firstname, ID, Email, Phone
- b. Prints the information for each employee in a nicely formatted layout. **DO NOT USE TABS**. Use this link as a guide for formatting layout.
<https://www.tutorialspoint.com/how-to-align-text-strings-using-python>
- c. The output will look like

LastName	FirstName	ID	Email	Phone
Harper	Larry	4066	Larry.Harper@email.edu	520-555-4066
Benton	Ryan	4071	Ryan.Benton@email.edu	520-555-4071
Farrell	Dave	4009	Dave.Farrell@email.edu	520-555-4009
Stone	Matthew	4041	Matthew.Stone@email.edu	520-555-4041
Martinez	Yessenia	4004	Yessenia.Martinez@email.edu	520-555-4004
Bernard	Mariah	4100	Mariah.Bernard@email.edu	520-555-4100
Day	Erin	4006	Erin.Day@email.edu	520-555-4006
Oconnell	Summer	4099	Summer.Oconnell@email.edu	520-555-4099
Stevens	Regina	4079	Regina.Stevens@email.edu	520-555-4079
Atkinson	Jonathon	4077	Jonathon.Atkinson@email.edu	520-555-4077
Levine	Sean	4026	Sean.Levine@email.edu	520-555-4026
Stephenson	Reginald	4083	Reginald.Stephenson@email.edu	520-555-4083
Garner	Kelli	4062	Kelli.Garner@email.edu	520-555-4062
Walsh	Kirk	4042	Kirk.Walsh@email.edu	520-555-4042

- iv. **Except for option 1**, your code should display the menu again after each display of information for the user to choose again. Therefore, you will need to use a while loop that exits when the choice is 1.

Notes:

1. All output data originate from the object lists or class variables
2. The functions defined above are not the only functions that you will write. You will write helper functions too.
3. Your constructors must have input validation
4. Your classes must have `__init__`, `__str__`, and `__repr__` functions
5. You should put all your code in one file named **project1Employees.py**. Normally, you would create a separate file for classes. However, due to the way D2L renames files, your code must be in one file.

Submission

Put all your code in one file named project1Employees.py.

Upload project1Employees.py to D2L

Do not upload the txt files. I will use my own.

Sample output for first project submission

Starting application...

Adding employees...

Added employee Larry Harper...

Added employee Ryan Benton...

Added employee Dave Farell...
etc...

Please select an option below

1. Quit
2. Display Employee Employment Information
3. Display Employee Contact Information

> 19

I am sorry, 19 is not an option.

Please select an option below

1. Quit
2. Display Employee Employment Information
3. Display Employee Contact Information

> 2

Employee Employment Information

LastName	FirstName	ID	Email	Phone	HireDate	Classification	Role	Salary
Harper	Larry	4066	Larry.Harper@email.edu	520-555-4066	5/5/2021	Full	Staff	65000.00
Benton	Ryan	4071	Ryan.Benton@email.edu	520-555-4071	10/6/2020	Full	Staff	56000.00
Farrell	Dave	4009	Dave.Farrell@email.edu	520-555-4009	10/6/2018	Part	Staff	34000.00
Stone	Matthew	4041	Matthew.Stone@email.edu	520-555-4041	8/23/2017	Full	Faculty	78000.00
Martinez	Yesenia	4004	Yesenia.Martinez@email.edu	520-555-4004	8/3/2017	Part	Staff	30000.00
Bernard	Mariah	4100	Mariah.Bernard@email.edu	520-555-4100	11/15/2016	Full	Faculty	120000.00
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Atkinson	Jonathon	4077	Jonathon.Atkinson@email.edu	520-555-4077	7/19/2011	Full	Faculty	48000.00
Levine	Sean	4026	Sean.Levine@email.edu	520-555-4026	10/9/2010	Part	Faculty	34000.00
Stephenson	Reginald	4083	Reginald.Stephenson@email.edu	520-555-4083	2/8/2010	Part	Staff	23000.00
Garner	Kelli	4062	Kelli.Garner@email.edu	520-555-4062	1/17/2010	Full	Faculty	58000.00
Walsh	Kirk	4042	Kirk.Walsh@email.edu	520-555-4042	4/8/2009	Full	Staff	92000.00

Please select an option below

1. Quit
2. Display Employee Employment Information
3. Display Employee Contact Information

> 3

Employee Contact Information

LastName	FirstName	ID	Email	Phone
Harper	Larry	4066	Larry.Harper@email.edu	520-555-4066
Benton	Ryan	4071	Ryan.Benton@email.edu	520-555-4071
Farrell	Dave	4009	Dave.Farrell@email.edu	520-555-4009
Stone	Matthew	4041	Matthew.Stone@email.edu	520-555-4041
Martinez	Yesenia	4004	Yesenia.Martinez@email.edu	520-555-4004
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Day	Erin	4006	Erin.Day@email.edu	520-555-4006
Oconnell	Summer	4099	Summer.Oconnell@email.edu	520-555-4099
Stevens	Regina	4079	Regina.Stevens@email.edu	520-555-4079
Atkinson	Jonathon	4077	Jonathon.Atkinson@email.edu	520-555-4077
Levine	Sean	4026	Sean.Levine@email.edu	520-555-4026
Stephenson	Reginald	4083	Reginald.Stephenson@email.edu	520-555-4083
Garner	Kelli	4062	Kelli.Garner@email.edu	520-555-4062
Walsh	Kirk	4042	Kirk.Walsh@email.edu	520-555-4042

Please select an option below

Please select an option below

1. Quit
2. Display Employee Employment Information
3. Display Employee Contact Information

> 1

Thank you for using the system.

Now exiting the program...

References

The data files to read in are included separately. The file contents are printed below for your reference.

employees.txt

LastName	FirstName	ID	Email	Phone	HireDate	Classification	Role	Salary
Harper	Larry	4066	Larry.Harper@email.edu	520-555-4066	5/5/2021	Full	Staff	65000.00
Benton	Ryan	4071	Ryan.Benton@email.edu	520-555-4071	10/6/2020	Full	Staff	56000.00
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Atkinson	Jonathon	4077	Jonathon.Atkinson@email.edu	520-555-4077	7/19/2011	Full	Faculty	48000.00
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Garner	Kelli	4062	Kelli.Garner@email.edu	520-555-4062	1/17/2010	Full	Faculty	58000.00
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